CSDF2

```python
import random
import string
from captcha.image import ImageCaptcha
from PIL import Image

# Generate random CAPTCHA text
characters = string.ascii_uppercase + string.digits
captcha_text = ''.join(random.choices(characters, k=6))

# Create CAPTCHA image
captcha = ImageCaptcha(width=250, height=80)
captcha.write(captcha_text, 'captcha.png')

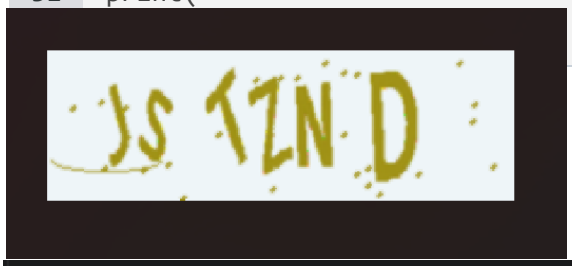
print("===== CAPTCHA GENERATION AND VERIFICATION =====")
print("CAPTCHA image generated: captcha.png")

# Open the generated image
Image.open('captcha.png').show()

# Verify user input
user_input = input("Enter the CAPTCHA shown in the image: ")

if user_input.upper() == captcha_text:
    print("CAPTCHA verified successfully!")
    print("Access Granted.")
else:
    print("Incorrect CAPTCHA!")
    print("Access Denied.")

print("================================================")
```




```
PS D:\submission\LP4> python -u "d:\submission\LP4\tempCodeRunnerFile.py"
===== CAPTCHA GENERATION AND VERIFICATION =====
CAPTCHA image generated: captcha.png
Enter the CAPTCHA shown in the image: 1JUOP
Incorrect CAPTCHA!
Access Denied.
================================================
PS D:\submission\LP4> python -u "d:\submission\LP4\tempCodeRunnerFile.py"
===== CAPTCHA GENERATION AND VERIFICATION =====
CAPTCHA image generated: captcha.png
Enter the CAPTCHA shown in the image: JSTZND
CAPTCHA verified successfully!
Access Granted.
================================================
PS D:\submission\LP4>
```